

A Rank-One Counterexample to the Domain-Free Cesàro/Maximum/Anti-Maximum Equivalence

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Lane 6 — candidate for human review

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Status. Candidate counterexample, likely valid. The result completely disproves the literal extension of source Theorem 5.2 obtained by deleting $\text{dom}(A) \subseteq E_u$. It does not completely resolve the source’s broader program of characterizing all relations between resolvent and Cesàro positivity.

1 Source question and exact scope

Arora and Glück’s Theorem 5.2 gives, under $\text{dom}(A) \subseteq E_u$, a four-way equivalence among eventual strong positivity of Cesàro means, strong positivity of the spectral projection, an individual strong maximum principle, and an individual strong anti-maximum principle [1, Theorem 5.2]. On PDF page 18 they ask for a better understanding of the relation between resolvent and Cesàro positivity when the domain inclusion is not imposed.

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- (iv) *The individual strong anti-maximum principle with respect to u holds, i.e., for every $0 \leq f \in E$, we have*

$$\mathcal{R}(\mu, A)f \preceq -u$$

for all μ in an f -dependent left neighbourhood of λ_0 .

Proof. By replacing A by $A - s(A)$, we may assume that $s(A) = 0$. The equivalence of (ii), (iii), and (iv) under the assumption $\text{dom}(A) \subseteq E_u$ was proved in [14, Theorem 4.4]. On the other hand, boundedness of the semigroup implies that 0 is actually a simple pole of the resolvent $\mathcal{R}(\cdot, A)$. We infer that $(e^{tA})_{t \geq 0}$ is uniformly mean ergodic and the corresponding mean ergodic projection coincides with the spectral projection P associated with 0; see [18, Theorem V.4.10]. Since $\text{dom}(A) \subseteq E_u$, the implication (ii) $\Rightarrow$ (i) now holds due to Proposition 5.1.

Lastly, if (i) holds, then so does (ii) by [14, Lemma 4.8(ii)]. $\square$

It has recently been shown by the present authors [6, Theorem 1.2] that the condition $\text{dom}(A) \subseteq E_u$ is, under mild assumptions on A , necessary for assertions (iii) and (iv) to hold simultaneously.

Clearly, the above results can only be considered as a starting point for a more thorough analysis, in particular since the methods used in this section are very closely related to the methods used in earlier articles about eventual positivity. In the future, it would, for instance, be desirable to better understand the relation between eventual positivity of the resolvent and the Cesàro means, without imposing the a priori assumption $\text{dom}(A) \subseteq E_u$. However, this seems to require more sophisticated analysis and probably different techniques.

We end this section with the observation that, by using Theorem 5.2, it can be shown for several semigroups considered in [7, 14, 15], that the Cesàro means are individually eventually positive.

Acknowledgements. The authors are grateful to Delio Mugnolo for pointing out the reference [30].

Figure 1: Source page 18: the end of Theorem 5.2 and the future-work direction concerning removal of $\text{dom}(A) \subseteq E_u$.

The precise question settled here is the sharp literal one: can the domain hypothesis simply be deleted while retaining the four-way equivalence? The answer is no, even for a bounded

generator of a positive semigroup. In fact, conditions (i)–(iii) of the source theorem hold and condition (iv) fails.

2 Definitions

Let E be a complex Banach lattice and $0 < u \in E$. Its principal ideal is

$$E_u = \{g \in E : |g| \leq cu \text{ for some } c \geq 0\}.$$

For real vectors g, h , write $g \succeq h$ if $g \geq ch$ for some $c > 0$. A positive u is quasi-interior when E_u is dense in E .

For a real C_0 -semigroup $(e^{tA})_{t \geq 0}$, its Cesàro mean at $r > 0$ is

$$C(r) = \frac{1}{r} \int_0^r e^{tA} dt.$$

The means are individually eventually strongly positive with respect to u if, for every $0 < f \in E$, one has $C(r)f \succeq u$ for all sufficiently large r . At a real pole λ_0 , the individual strong maximum and anti-maximum principles require the corresponding strong order inequalities for $\mathcal{R}(\lambda, A)f$ on a right, respectively left, f -dependent neighbourhood of λ_0 .

3 Idea of the proof

Use the averaging projection $Pf = (\int f)\mathbf{1}$ and put $A = P - I$. The semigroup and every Cesàro mean are convex combinations of I and P , so the positive rank-one part supplies an explicit lower bound by $\mathbf{1}$. The right resolvent has the same favourable sign pattern. On the left of the pole, however, the identity coefficient is positive while the rank-one coefficient is negative. A positive unbounded L^2 function therefore escapes every upper bound by a negative multiple of $\mathbf{1}$. This is exactly the regularity obstruction hidden by $\text{dom}(A) \subseteq E_u$.

4 Counterexample

Theorem 1. *Let $E = L^2(0, 1; \mathbb{C})$ with its usual Banach-lattice order, let $u = \mathbf{1}$, and define*

$$\varphi(f) = \int_0^1 f(s) ds, \quad Pf = \varphi(f)\mathbf{1}, \quad A = P - I.$$

Then:

- (i) *A generates a bounded positive uniformly continuous semigroup, $s(A) = 0$ is a simple pole of the resolvent, and its spectral projection is P ;*
- (ii) *the Cesàro means are strongly positive with respect to u for every $r > 0$;*
- (iii) *$Pf \succeq u$ for every $0 < f \in E$;*
- (iv) *the individual strong maximum principle holds at 0, in fact on the entire interval $(0, \infty)$;*
- (v) *the individual strong anti-maximum principle at 0 fails.*

Moreover, $\text{dom}(A) = E \not\subseteq E_u$. Hence assertions (i)–(iii) of Arora–Glück Theorem 5.2 hold while assertion (iv) fails after the domain hypothesis is removed.

Proof. Since $\varphi(\mathbf{1}) = 1$, the positive rank-one operator P is a projection. Also $u = \mathbf{1}$ is quasi-interior and

$$E_u = L^\infty(0, 1) \subsetneq L^2(0, 1) = \text{dom}(A).$$

Because P and $I - P$ are complementary projections,

$$A = 0 \cdot P - 1 \cdot (I - P), \quad \sigma(A) = \{0, -1\}.$$

Thus $0 = s(A)$ is a simple pole and P is its spectral projection.

Functional calculus for the two complementary projections gives

$$e^{tA} = P + e^{-t}(I - P) = e^{-t}I + (1 - e^{-t})P. \quad (1)$$

This is positive for every $t \geq 0$ and bounded uniformly in t . Integrating (1), we obtain

$$C(r) = a_r I + (1 - a_r)P, \quad a_r = \frac{1 - e^{-r}}{r} \in (0, 1) \quad (2)$$

for every $r > 0$. If $0 < f \in E$, strict positivity of integration on L_+^2 gives $\varphi(f) > 0$. Hence

$$C(r)f \geq (1 - a_r)\varphi(f)\mathbf{1} \succeq \mathbf{1}, \quad Pf = \varphi(f)\mathbf{1} \succeq \mathbf{1}. \quad (3)$$

This proves the Cesàro and spectral assertions. Notice that the lower bound in (3) can even be made uniform for all $r \geq r_0 > 0$.

For $\lambda \notin \{-1, 0\}$, decomposition along P and $I - P$ yields

$$\begin{aligned} \mathcal{R}(\lambda, A) &= \frac{1}{\lambda}P + \frac{1}{\lambda + 1}(I - P) \\ &= \frac{1}{\lambda + 1}I + \frac{1}{\lambda(\lambda + 1)}P. \end{aligned} \quad (4)$$

If $\lambda > 0$ and $0 < f \in E$, both coefficients in the last line are positive, and therefore

$$\mathcal{R}(\lambda, A)f \geq \frac{\varphi(f)}{\lambda(\lambda + 1)}\mathbf{1}.$$

This proves the individual strong maximum principle (and its usual scaled form uniformly on every interval $(0, \varepsilon)$).

Now take the fixed positive vector

$$f(x) = x^{-1/4} \in L^2(0, 1), \quad \varphi(f) = \frac{4}{3}.$$

For every $\mu \in (-1, 0)$, (4) gives

$$\mathcal{R}(\mu, A)f(x) = \frac{1}{1 + \mu} \left(x^{-1/4} - \frac{4}{3|\mu|} \right). \quad (5)$$

The right-hand side tends to $+\infty$ as $x \downarrow 0$. In particular it is positive on a set of positive measure, so it is not ≤ 0 and a fortiori cannot satisfy $\mathcal{R}(\mu, A)f \preceq -\mathbf{1}$. The same fixed f fails for every $\mu \in (-1, 0)$, which rules out every left neighbourhood of the pole. This proves the anti-maximum failure and completes the proof. $\square$

Corollary 1. *The domain hypothesis in source Theorem 5.2 cannot be omitted, even if one strengthens the ambient assumptions by requiring A to be bounded and the semigroup to be positive for every time.*

5 Audit of the previously printed rank-one example

The same operator on $L^2(0, 1)$ appears on PDF page 3 of Arora–Glück, arXiv:2203.05680v2 [2]. The displayed resolvent formula there agrees with (4), but the following operator anti-maximum inequality asserted in that passage does not.

Let us briefly discuss the particular case where $E = C(K)$, the space of continuous functions on a compact Hausdorff set K and $u \in E$ is the constant one function. Since $E_u = E$, the domination condition is always satisfied in Setting 1.1 for $n = 1$. Thus, according to Theorem 1.2, both strong individual maximum and anti-maximum principles hold at λ_0 if and only if the spectral assumption is satisfied.

Next, let $E = L^p(\Omega, \mu)$, where (Ω, μ) is a finite measure space and $p \in [1, \infty)$. If u is the constant one function, then as mentioned before, $E_u = L^\infty(\Omega, \mu)$. For numerous differential operators A , one can show using Sobolev embedding theorems that the domination condition holds for sufficiently large $n \in \mathbb{N}$ but not for $n = 1$. As a result, if one knows a priori that the maximum principle holds at λ_0 for those operators that in addition satisfy the spectral assumption, then the anti-maximum principle cannot hold at λ_0 . Analogously, if the anti-maximum principle holds at λ_0 , then the maximum principle cannot. We will illustrate this in Section 5.

It is natural to ask whether the assertions in Theorem 1.2 are equivalent without the domination assumption in Setting 1.1. However, one can immediately check that this is not the case. Indeed, on $L^2[0, 1]$ consider the bounded operator defined by $Af = (\mathbb{1} \otimes \mathbb{1})f - f$ for all $f \in L^2[0, 1]$, and let $\lambda_0 = 0$. The resolvent can be computed directly and is given by

$$\mathcal{R}(\lambda, A)f = \frac{1}{\lambda(\lambda + 1)}((\mathbb{1} \otimes \mathbb{1})f + \lambda f), \quad (\lambda \in \rho(A)).$$

Thus, one has $\mathcal{R}(\lambda, A) \succeq \mathbb{1} \otimes \mathbb{1}$ and $\mathcal{R}(\mu, A) \preceq -\mathbb{1} \otimes \mathbb{1}$ for all $\lambda > 0$ and for all $\mu \in (-1, 0)$. However, $\text{dom}(A^n) = L^2[0, 1] \not\subseteq L^\infty[0, 1] = L^2[0, 1]_1$ for any $n \in \mathbb{N}_0$.

Related literature. Individual maximum and anti-maximum principles for concrete differential operators have been intensively studied for several decades. For instance, we refer the reader to the articles [12, 13, 24, 31, 32] and to the references in Section 5 for various examples. Abstract approaches to (anti-)maximum principles with several applications have been taken [8, 19, 20, 36]. Recently, the first author

Figure 2: Supporting-paper page 3: the rank-one passage under audit.

Indeed, at $\mu = -\frac{1}{2}$, formula (4) becomes

$$\mathcal{R}(-\frac{1}{2}, A) = 2I - 4P.$$

For the vector $f(x) = x^{-1/4}$ used above,

$$\mathcal{R}(-\frac{1}{2}, A)f = 2x^{-1/4} - \frac{16}{3},$$

which is positive whenever $0 < x < (3/8)^4$. Thus this vector disproves even ordinary anti-maximum negativity. The issue is confined to this auxiliary example: Theorem 1.2 of that paper assumes a weaker domination condition $\text{dom}(A^n) \subseteq E_u$ for some n , whereas the bounded operator here has $\text{dom}(A^n) = L^2(0, 1) \not\subseteq L^\infty(0, 1)$ for every $n \geq 0$.

6 Verification and limitations

The proof was checked independently in four decomposed steps: projection and spectrum, semigroup/Cesàro formula, resolvent formula, and the single-vector order obstruction. At $\mu = -1/2$ the calculation reduces to the especially transparent identity $2I - 4P$. All conclusions are exact and require no numerical experiment.

The bounded novelty search covered the local run indexes and full-source corpus, the two cited arXiv records, exact title/id searches, close keyword searches, and erratum/correction searches. No later correction or exact statement of this packet’s conclusion was found as of 9 August 2026. The construction itself is printed in [2]; the candidate-new content is the corrected order analysis and its use against the domain-free extension of [1, Theorem 5.2].

This is not a full characterization of the broad future-work direction. In this example both the Cesàro means and the right-hand resolvent are strongly positive, so it does not separate those two properties. It proves the sharp negative statement that those properties and strong spectral positivity do not recover the anti-maximum conclusion without domain regularity.

Human review focus. Check (5), the order quantifiers on the whole interval $(-1, 0)$, and the scope of the literature-audit statement. The candidate mathematical result is recommended for promotion after those checks.

References

- [1] S. Arora and J. Glück, *Criteria for eventual domination of operator semigroups and resolvents*, Operator Theory: Advances and Applications 292 (2023), 1–26; [arXiv:2204.00146v3](#).
- [2] S. Arora and J. Glück, *A characterization of the individual maximum and anti-maximum principle*, Math. Z. 305 (2023), article 49; [arXiv:2203.05680v2](#).